Project 36: Startup Financial Integrity Checker

AI-driven anomaly detection for validating accounting and financial integrity.

Project Overview

Problem	Fraudulent accounting entries and hidden manipulations often escape manual audits.
Solution	AI anomaly detection system that identifies inconsistent ledgers, suspicious entries, and cash flow mismatches.
AI Technique	Machine Learning based anomaly detection on structured accounting datasets.
Tech Stack	Pandas, SQL, Python
Outcome	Prevents financial fraud and ensures trustable reporting for investors.

Mock Financial Integrity Dataset

Startup Name	Revenue (₹Cr)	Expense (₹Cr)	Cashflow Match %	Anomaly Score (0–100)	Integrity Rating	AI Remark
FinServe AI	120	96	97%	**92**	Excellent	Stable financial records, minimal variance in monthly flows.
EcoBuild Solutions	75	70	89%	**77**	Moderate	Detected inconsistent vendor payments in Q2; review suggested.
HealthNova Pvt Ltd	150	140	72%	**48**	Poor	Unexplained cash deposits and round-off manipulations flagged.
AgroTech India	60	55	95%	**88**	Strong	All ledger entries consistent with banking data.

AI Audit Scoring Breakdown

Metric	Weight (%)	FinServe	EcoBuild	HealthNova	AgroTech	Justification
Transaction Consistency	35	95	80	50	90	Cross-check between ledger and bank statements.
Expense Correlation	25	90	75	55	88	AI correlation between declared expenses and cost drivers.
Cashflow Stability	25	94	78	40	89	Deviation of monthly inflow vs outflow trends.
Debt-to-Asset Health	15	85	74	47	91	AI analysis of leverage levels vs revenue stability.

Final Financial Integrity Index

Startup	Integrity Index (0–100)	Risk Level
FinServe AI	**92**	Low Risk
EcoBuild Solutions	**77**	Moderate Risk
HealthNova Pvt Ltd	**48**	High Risk
AgroTech India	**88**	Low Risk

Impact Summary

- =Early detection of hidden frauds and inconsistencies in financial reporting.
- >**AI-based anomaly scoring** brings predictive audit intelligence.
- =u Strengthens investor trust and compliance transparency.
- =Reduces financial losses due to data manipulation or oversight.